Online Appendix — The Transmission Efficiency of Net Investment Is Decreasing in the Replacement Burden

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This appendix supports "The Transmission Efficiency of Net Investment Is Decreasing in the Replacement Burden." Section and table references of the form "Section 4" or "Table 3" point to the main text; tables here are numbered OA1, OA2, OA3. It contains the full variable construction (OA.1), the inference comparison that motivates estimating the country gradient on value-added-free denominators (OA.2), the firm-level clustering sensitivity (OA.3), and the complete firm gradient summarized in main-text Table 3 (OA.4). It deliberately contains nothing on the determinants of δ — what drives the replacement burden, or whether it is rising — which lie outside the scope of this paper (Section 1); the appendix, like the paper, claims only the shape of the transmission relationship.

OA.1 Data and Variable Construction

Country panel (value-added-free, primary). Penn World Table 11.0 (Feenstra, Inklaar, and Timmer 2015), 31 OECD economies, 1970–2023, 1,643 country-years. The replacement burden is measured value-added-free as $\delta_K = \text{CFC}/K$, taken from the PWT depreciation series; net investment is the growth rate of the real net capital stock, $n_K = \Delta K/K$ (variable `rnna`); the outcome is labour-productivity growth, $\Delta \ln(\text{rgdpna/emp})$. Value added enters none of the three: not the splitter, not the regressor, not the regressor's denominator. The regime cut τ is swept across percentiles of the δ_K distribution.

Country panel (conventional output denominator). OECD STAN, 32 countries, with $\delta = \text{CFC/VA}$, a net-investment-to-value-added rate, and value-added growth. This panel is used only as the comparison case in OA.2 and as the target of the placebo calibration (main-text Figure A1); it is not the basis of any reported gradient.

Firm panel. Compustat North America Annual, 1975–2023. Financial firms (SIC 6000–6999) and regulated utilities (SIC 4900–4999) are excluded; observations require positive total assets, positive net property, plant and equipment, and non-negative depreciation. The replacement burden is $\delta = \text{DP/PPENT}$ (depreciation over net PPE); net investment is $n = (\text{CAPX} - \text{DP})/\text{PPENT}_{t-1}$; the outcome is asset growth. The resulting panel is 251,130 firm-years on 23,955 firms. All ratios are winsorized at the 1st and 99th percentiles. Firm fixed effects are absorbed by the within transformation; standard errors are clustered by firm. Because the firm denominators are PPE rather than value added, the shared-output channel of Section 2.2 cannot operate at this level.

Inference. With 31–32 country clusters the asymptotic clustered standard error is unreliable, so country significance is assessed by the wild cluster bootstrap (Rademacher weights, restricted null $\gamma_{\text{above}} = \gamma_{\text{below}}$, 399 replications) on the regime gap. With roughly 24,000 firm clusters the few-cluster problem does not arise; firm significance is read from clustered standard errors, and OA.3 reports its sensitivity to the clustering dimension.

OA.2 Why the Country Gradient Is Estimated on Value-Added-Free Denominators

The main text estimates the country gradient on value-added-free denominators (Section 4) and notes in passing that the conventional output-denominator specification yields a similar point-estimate gradient whose precision does not survive few-cluster-robust inference (Section 2.2). Table OA1 gives that comparison in full. For each specification it reports the attenuation, the p-value on the regime gap under naive clustered standard errors, and the p-value under the wild cluster bootstrap.

Table OA1: Country gradient — naive clustered inference versus the wild cluster bootstrap, for the value-added-free specification (main result) and the conventional output-denominator specification. 31–32 country clusters.

Specification	Cutoff	Attenuation	p (naive cluster)	p (wild bootstrap)
Value-added-free (δ_K, $\Delta K/K$)	40th pctile	25%	< 0.001	0.010
	55th pctile	28%	< 0.001	0.003
	70th pctile	33%	< 0.001	0.003
	80th pctile	18%	< 0.001	0.160
	90th pctile	14%	< 0.001	0.150
Conventional ($\delta =$ CFC/VA)	$\tau = 18.5\%$	42%	< 0.001	0.037
	$\tau = 20.0\%$	41%	0.002	0.158
	$\tau = 21.0\%$	35%	0.005	0.235
	$\tau = 23.4\%$	47%	0.047	0.330

Two things are visible. First, naive clustered standard errors report near-certain significance for both specifications — the conventional gradient looks as decisive as the clean one. Second, the wild cluster bootstrap separates them: the value-added-free gradient remains significant through the bulk of the distribution ($p \leq 0.01$ at the 40th–70th percentiles), while the conventional gradient is significant only at its lowest cutoff and is otherwise indistinguishable from zero (p between 0.16 and 0.33). The apparent precision of the conventional specification is largely an artifact of the asymptotic standard error; once inference is made robust to the small number of clusters, the support that remains is on the value-added-free denominators — where, by construction, the shared-output channel cannot have produced it. This is the empirical reason the main result is estimated value-added-free rather than a stylistic preference, and it is why the conventional specification is not reported as a result in its own right.

OA.3 Firm-Level Clustering Sensitivity

At the firm level the concern is the opposite of the country case: with so many observations, precision is abundant, and the question is whether it survives progressively more conservative treatment of within-firm and within-year dependence. Table OA2 re-estimates the high-δ coefficient γ_{above} at two representative thresholds under heteroskedasticity-robust standard errors, standard errors clustered by firm, and two-way standard errors clustered by firm and by year.

Table OA2: Firm high-δ transmission coefficient γ_{above} and its t-statistic under three error structures. Firm fixed effects throughout.

δ threshold	γ_{above}	t (robust)	t (cluster: firm)	t (two-way: firm $\times$ year)
0.20	15.9	16.2	15.0	5.9
0.30	7.4	6.8	6.5	3.0

Two-way clustering by firm and year is the most demanding standard treatment, allowing arbitrary correlation across firms within a year and across years within a firm; it roughly halves the t-statistic relative to firm-only clustering. The coefficient remains significant under it at both thresholds ($t = 5.9$ and $t = 3.0$). Since the low-δ coefficient $\gamma_{\text{below}} \approx 50$ is estimated still more precisely, the regime gap that defines the gradient is significant under every error structure considered.

OA.4 Full Firm Gradient

Table OA3 is the complete version of main-text Table 3: both regime coefficients, the attenuation, the p-value on the high-δ coefficient, and the number of high-δ firm-years, across the threshold sweep.

Table OA3: Firm-level attenuation gradient, full detail (Compustat, asset growth). 23,955 firms, 251,130 firm-years. Firm FE, SE clustered by firm.

δ threshold (DP/PPE)	γ_{below}	γ_{above}	Attenuation	p (γ_{above})	N_{above}
0.10	51.7	26.9	48%	< 0.001	196,910
0.20	52.1	15.9	69%	< 0.001	110,387
0.30	51.6	7.4	86%	< 0.001	71,429
0.50	50.4	0.1	99.8%	0.94	37,468
1.00	46.1	2.4	95%	0.03	14,816

The low-δ coefficient is stable near 50 across thresholds; the high-δ coefficient falls monotonically, with attenuation rising from 48% at $\delta = 0.10$ to 86% at $\delta = 0.30$, each precisely estimated. Beyond $\delta \approx 0.30$ the high-δ coefficient is statistically indistinguishable from zero — transmission has effectively vanished — though the point estimates in the sparse high tail are correspondingly imprecise, consistent with the main text's decision (Section 2.4) to claim the onset of attenuation rather than a terminal value.

References

Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2015). The next generation of the Penn World Table. *American Economic Review*, 105(10), 3150–3182.